

Task1

```
srinivas@Srinivas:~$ mkdir linux_task1
srinivas@Srinivas:~$ ls
CPrograms  a.out  d1  f1.txt  fork.c  fork3.c  hello.c  linux_task1  ss  subsystemcalls  wait.c  waitfork.c
srinivas@Srinivas:~$ cd linux_task1
srinivas@Srinivas:~/linux_task1$ touch file1.txt file2.txt file3.txt file4.txt file5.txt
srinivas@Srinivas:~/linux_task1$ is
Command 'is' not found, but can be installed with:
sudo apt install ironseed
srinivas@Srinivas:~/linux_task1$ ls
file1.txt  file2.txt  file3.txt  file4.txt  file5.txt
srinivas@Srinivas:~/linux_task1$ ls -l
total 0
-rw-r--r-- 1 srinivas srinivas 0 Aug  3 14:29 file1.txt
-rw-r--r-- 1 srinivas srinivas 0 Aug  3 14:29 file2.txt
-rw-r--r-- 1 srinivas srinivas 0 Aug  3 14:29 file3.txt
-rw-r--r-- 1 srinivas srinivas 0 Aug  3 14:29 file4.txt
-rw-r--r-- 1 srinivas srinivas 0 Aug  3 14:29 file5.txt
srinivas@Srinivas:~/linux_task1$ echo "Hello! Welcome to Operating System" > file1.txt
srinivas@Srinivas:~/linux_task1$ cat file1.txt
Hello! Welcome to Operating System
srinivas@Srinivas:~/linux_task1$ rm file1.txt
srinivas@Srinivas:~/linux_task1$ ls
file2.txt  file3.txt  file4.txt  file5.txt
srinivas@Srinivas:~/linux_task1$ rm file5.txt
srinivas@Srinivas:~/linux_task1$ ls
file2.txt  file3.txt  file4.txt
srinivas@Srinivas:~/linux_task1$ ls -l
total 0
-rw-r--r-- 1 srinivas srinivas 0 Aug  3 14:29 file2.txt
-rw-r--r-- 1 srinivas srinivas 0 Aug  3 14:29 file3.txt
-rw-r--r-- 1 srinivas srinivas 0 Aug  3 14:29 file4.txt
srinivas@Srinivas:~/linux_task1$ |
```

Task2

```

srinivas@Srinivas:~$ mkdir linux_task2
mkdir: linux_task2: File exists
srinivas@Srinivas:~$ ls
CPrograms  fl.txt    hello.c    ss          waitfork.c
a.out      fork.c    linux_task1 subsystemcalls
d1         fork3.c   linux_task2 wait.c
srinivas@Srinivas:~$ ls -l
total 60
drwxr-xr-x 2 srinivas srinivas 4096 Jul 29 08:42 CPrograms
-rwxr-xr-x 1 srinivas srinivas 16080 Aug 1 06:07 a.out
drwxr-xr-x 2 srinivas srinivas 4096 Jul 29 07:07 d1
-rwxrwxrwx 1 srinivas srinivas 0 Aug 1 05:02 fl.txt
-rw-r--r-- 1 srinivas srinivas 388 Aug 1 06:06 fork.c
-rw-r--r-- 1 srinivas srinivas 389 Aug 1 05:49 fork3.c
-rw-r--r-- 1 srinivas srinivas 147 Jul 29 08:05 hello.c
drwxr-xr-x 3 srinivas srinivas 4096 Aug 3 14:58 linux_task1
drwxr-xr-x 2 srinivas srinivas 4096 Aug 3 14:53 linux_task2
drwxr-xr-x 2 srinivas srinivas 4096 Jul 29 08:20 ss
drwxr-xr-x 2 srinivas srinivas 4096 Aug 1 04:35 subsystemcalls
-rw-r--r-- 1 srinivas srinivas 18 Aug 1 04:51 wait.c
-rw-r--r-- 1 srinivas srinivas 389 Aug 1 05:58 waitfork.c
srinivas@Srinivas:~$ cd linux_task2
srinivas@Srinivas:~/linux_task2$ touch source.txt notes.txt data.txt
srinivas@Srinivas:~/linux_task2$ ls
data.txt  notes.txt  source.txt
srinivas@Srinivas:~/linux_task2$ ls-l
ls-l: command not found
srinivas@Srinivas:~/linux_task2$ ls -l
total 0
-rw-r--r-- 1 srinivas srinivas 0 Aug 4 02:55 data.txt
-rw-r--r-- 1 srinivas srinivas 0 Aug 4 02:55 notes.txt
-rw-r--r-- 1 srinivas srinivas 0 Aug 4 02:55 source.txt
srinivas@Srinivas:~/linux_task2$ echo "this is the source file." > source.txt
srinivas@Srinivas:~/linux_task2$ cat source.txt
this is the source file.
srinivas@Srinivas:~/linux_task2$ mkdir backup
srinivas@Srinivas:~/linux_task2$ cp source.txt backup/
srinivas@Srinivas:~/linux_task2$ cp data.txt backup/data_backup.txt
srinivas@Srinivas:~/linux_task2$ mv notes.txt backup/
srinivas@Srinivas:~/linux_task2$ ls
backup  data.txt  source.txt

```

```

srinivas@Srinivas:~$ cd linux_task2
srinivas@Srinivas:~/linux_task2$ cd backup
srinivas@Srinivas:~/linux_task2/backup$ ls
data_backup.txt  notes.txt  source.txt
srinivas@Srinivas:~/linux_task2/backup$ cd linux_task2
-bash: cd: linux_task2: No such file or directory
srinivas@Srinivas:~/linux_task2/backup$ cd ..
srinivas@Srinivas:~/linux_task2$ mv source.txt original.txt
srinivas@Srinivas:~/linux_task2$ ls -R
.:
backup  data.txt  original.txt

./backup:
data_backup.txt  notes.txt  source.txt

```

```

srinivas@Srinivas:~$ mkdir linux_task3
mkdir: linux_task3: File exists
srinivas@Srinivas:~$ ls
CPrograms  fl.txt  hello.c      linux_task3  wait.c
a.out      fork.c  linux_task1  ss           waitfork.c
d1         fork3.c linux_task2  subsystemcalls
srinivas@Srinivas:~$ ls -l
total 64
drwxr-xr-x 2 srinivas srinivas 4096 Jul 29 08:42 CPrograms
-rwxr-xr-x 1 srinivas srinivas 16080 Aug 1 06:07 a.out
drwxr-xr-x 2 srinivas srinivas 4096 Jul 29 07:07 d1
-rwxrwxrwx 1 srinivas srinivas 0 Aug 1 05:02 fl.txt
-rw-r--r-- 1 srinivas srinivas 388 Aug 1 06:06 fork.c
-rw-r--r-- 1 srinivas srinivas 389 Aug 1 05:49 fork3.c
-rw-r--r-- 1 srinivas srinivas 147 Jul 29 08:05 hello.c
drwxr-xr-x 3 srinivas srinivas 4096 Aug 3 14:58 linux_task1
drwxr-xr-x 3 srinivas srinivas 4096 Aug 4 03:01 linux_task2
drwxr-xr-x 2 srinivas srinivas 4096 Aug 4 03:05 linux_task3
drwxr-xr-x 2 srinivas srinivas 4096 Jul 29 08:20 ss
drwxr-xr-x 2 srinivas srinivas 4096 Aug 1 04:35 subsystemcalls
-rw-r--r-- 1 srinivas srinivas 18 Aug 1 04:51 wait.c
-rw-r--r-- 1 srinivas srinivas 389 Aug 1 05:58 waitfork.c
srinivas@Srinivas:~$ cd linux_task3
srinivas@Srinivas:~/linux_task3$ touch students.txt
srinivas@Srinivas:~/linux_task3$ ls
students.txt
srinivas@Srinivas:~/linux_task3$ cat > students.txt
s1
s2
s3
s4
s5
s6
s7
s8

s9
s0
s1
s2
s3
s4
s5

```

```
s0
srinivas@Srinivas:~/linux_task3$ cat students.txt
s1
s2
s3
s4
s5
s6
s7
s8

s9
s0
s1
s2
s3
s4
s5
srinivas@Srinivas:~/linux_task3$ head -5 students.txt
s1
s2
s3
s4
s5
srinivas@Srinivas:~/linux_task3$ tail -5 students.txt
s1
s2
s3
s4
s5
srinivas@Srinivas:~/linux_task3$ head -10 student.txt
head: cannot open 'student.txt' for reading: No such file or directory
srinivas@Srinivas:~/linux_task3$ head -10 students.txt
s1
s2
s3
s4
s5
s6
s7
s8

s9
```

```
srinivas@Srinivas:~/linux_task3$ tail -10 students.txt
s7
s8

s9
s0
s1
s2
s3
s4
s5
```

Task4

```
srinivas@Srinivas:~$ mkdir organization
srinivas@Srinivas:~$ ls
CPrograms  fl.txt  hello.c      linux_task3  ss          waitfork.c
a.out      fork.c   linux_task1  oranization  subsystemcalls
d1         fork3.c  linux_task2  organization wait.c
srinivas@Srinivas:~$ cd organization
srinivas@Srinivas:~/organization$ touch file.txt
srinivas@Srinivas:~/organization$ touch file1.txt file2.txt file3.txt image1.jpg image2.jpg
srinivas@Srinivas:~/organization$ ls
file.txt  file1.txt  file2.txt  file3.txt  image1.jpg  image2.jpg
srinivas@Srinivas:~/organization$ mkdir documents
srinivas@Srinivas:~/organization$ mkdir images
srinivas@Srinivas:~/organization$ mv *.txt documents/
srinivas@Srinivas:~/organization$ ls
documents  image1.jpg  image2.jpg  images
srinivas@Srinivas:~/organization$ mv *.jpg images/
srinivas@Srinivas:~/organization$ ls
documents  images
srinivas@Srinivas:~/organization$ ls documents
file.txt  file1.txt  file2.txt  file3.txt
srinivas@Srinivas:~/organization$ ls images
image1.jpg  image2.jpg
srinivas@Srinivas:~/organization$ mv documents/file1.txt documents/notes.txt
srinivas@Srinivas:~/organization$ ls documents
file.txt  file2.txt  file3.txt  notes.txt
srinivas@Srinivas:~/organization$ mv images/image1.jpg images/photo.jpg
srinivas@Srinivas:~/organization$ ls
documents  images
srinivas@Srinivas:~/organization$ ls images
image2.jpg  photo.jpg
srinivas@Srinivas:~/organization$ ls -R
.:
documents  images

./documents:
file.txt  file2.txt  file3.txt  notes.txt

./images:
image2.jpg  photo.jpg
srinivas@Srinivas:~/organization$ cd ..
srinivas@Srinivas:~$ |
```

Task5

```

srinivas@Srinivas:~$ mkdir project
mkdir: project: File exists
srinivas@Srinivas:~$ ls
CPrograms  fl1.txt  hello.c      linux_task3  project      wait.c
a.out      fork.c    linux_task1  oranization  ss           waitfork.c
d1         fork3.c   linux_task2  organization subsystemcalls
srinivas@Srinivas:~$ cd project
srinivas@Srinivas:~/project$ mkdir -p source/c source/python docs backup
srinivas@Srinivas:~/project$ ls
backup  docs  source
srinivas@Srinivas:~/project$ touch source/c/file1.c source/c/file2.c source/c/file3.c
srinivas@Srinivas:~/project$ touch source/python/script1.py source/python/script2.py
srinivas@Srinivas:~/project$ ls -R
.:
backup  docs  source

./backup:

./docs:

./source:
c  python

./source/c:
file1.c  file2.c  file3.c

./source/c/file1.c:

./source/c/file2.c:

./source/c/file3.c:

./source/python:
script1.py  script2.py
srinivas@Srinivas:~/project$ rmdir backup
srinivas@Srinivas:~/project$ ls
docs  source

```

```

srinivas@Srinivas:~/project$ mkdir backup
mkdir: backup: File exists
srinivas@Srinivas:~/project$ touch backup/backup.txt
srinivas@Srinivas:~/project$ ls
backup  docs  source
srinivas@Srinivas:~/project$ mkdir backup
mkdir: backup: File exists
srinivas@Srinivas:~/project$ touch backup/backup.txt
srinivas@Srinivas:~/project$ rmdir
error: the following required arguments were not provided:
  <dirs>...

Usage: rmdir [OPTION]... DIRECTORY...

For more information, try '--help'.
srinivas@Srinivas:~/project$ rmdir backup
rmdir: failed to remove 'backup': Directory not empty
srinivas@Srinivas:~/project$ rm backup/backup.txt
srinivas@Srinivas:~/project$ ls backup
srinivas@Srinivas:~/project$ ls backup/
srinivas@Srinivas:~/project$ rmdir backup
srinivas@Srinivas:~/project$ ls
docs  source
srinivas@Srinivas:~/project$ |

```

Task6

```
srinivas@Srinivas:~$ mkdir linux_task4
srinivas@Srinivas:~$ cd linux_task4
srinivas@Srinivas:~/linux_task4$ touch public.txt private.txt script.sh
srinivas@Srinivas:~/linux_task4$ echo "This is a public file." > public.txt
srinivas@Srinivas:~/linux_task4$ echo "this is a private file." > private.txt
srinivas@Srinivas:~/linux_task4$ echo "echo HELLO, linux!" > script.sh
srinivas@Srinivas:~/linux_task4$ ls -l
total 12
-rw-r--r-- 1 srinivas srinivas 24 Aug  4 04:53 private.txt
-rw-r--r-- 1 srinivas srinivas 23 Aug  4 04:52 public.txt
-rw-r--r-- 1 srinivas srinivas 19 Aug  4 04:53 script.sh
srinivas@Srinivas:~/linux_task4$ chmod 600 private.txt
srinivas@Srinivas:~/linux_task4$ chmod 644 public.txt
srinivas@Srinivas:~/linux_task4$ chmod +x script.sh
srinivas@Srinivas:~/linux_task4$ ls -l
total 12
-rw----- 1 srinivas srinivas 24 Aug  4 04:53 private.txt
-rw-r--r-- 1 srinivas srinivas 23 Aug  4 04:52 public.txt
-rwxr-xr-x 1 srinivas srinivas 19 Aug  4 04:53 script.sh
srinivas@Srinivas:~/linux_task4$ chmod -x script.sh
srinivas@Srinivas:~/linux_task4$ ls -l
total 12
-rw----- 1 srinivas srinivas 24 Aug  4 04:53 private.txt
-rw-r--r-- 1 srinivas srinivas 23 Aug  4 04:52 public.txt
-rw-r--r-- 1 srinivas srinivas 19 Aug  4 04:53 script.sh
srinivas@Srinivas:~/linux_task4$ |
```

Task7

```
srinivas@Srinivas:~$ mkdir perission_lab
mkdir: perission_lab: File exists
srinivas@Srinivas:~$ cd perission_lab
srinivas@Srinivas:~/perission_lab$ touch test.txt
srinivas@Srinivas:~/perission_lab$ echo "this is a linux perission test." > test.txt
srinivas@Srinivas:~/perission_lab$ chmod u-r test.txt
srinivas@Srinivas:~/perission_lab$ cat test.txt
cat: test.txt: Permission denied
srinivas@Srinivas:~/perission_lab$ chmod u+r test.txt
srinivas@Srinivas:~/perission_lab$ chmod u-w test.txt
srinivas@Srinivas:~/perission_lab$ echo "Adding another line." >> test.txt
-bash: test.txt: Permission denied
srinivas@Srinivas:~/perission_lab$ chmod u+w test.txt
srinivas@Srinivas:~/perission_lab$ chmod u-x test.txt
srinivas@Srinivas:~/perission_lab$ ./test.txt
-bash: ./test.txt: Permission denied
srinivas@Srinivas:~/perission_lab$ ls -l test.txt
-rw-r--r-- 1 srinivas srinivas 32 Aug  4 05:03 test.txt
srinivas@Srinivas:~/perission_lab$ |
```

Task 8

```
srinivas@Srinivas:~$ mkdir linux_task6
mkdir: linux_task6: File exists
srinivas@Srinivas:~$ ls
CPrograms  fork.c      linux_task2  oranization    ss
a.out      fork3.c     linux_task3  organization   subsystemcalls
dl         hello.c     linux_task4  perission_lab  wait.c
fl.txt     linux_task1 linux_task6  project        waitfork.c
srinivas@Srinivas:~$ cd linux_task6
srinivas@Srinivas:~/linux_task6$ touch important.txt
srinivas@Srinivas:~/linux_task6$ cat > important.txt
Line 1: Linux is powerful.
Line 2: File permissions are important.
Line 3: Backups prevent data loss.
Line 4: cp is used to copy files.
Line 5: mv is used to rename files.
Line 6: rm deletes files.
Line 7: Directories organize files.
Line 8: chmod changes permissions.
Line 9: Practice improves Linux skills.
Line 10: This is the last line.

srinivas@Srinivas:~/linux_task6$ mkdir backup
srinivas@Srinivas:~/linux_task6$ ls -R
.:
backup  important.txt

./backup:
srinivas@Srinivas:~/linux_task6$ cp important.txt backup/
srinivas@Srinivas:~/linux_task6$ mv backup/important.txt backup/important_backup.txt
srinivas@Srinivas:~/linux_task6$ rm important.txt
srinivas@Srinivas:~/linux_task6$ cp backup/important_backup.txt .
srinivas@Srinivas:~/linux_task6$ ls -R
.:
backup  important_backup.txt

./backup:
important_backup.txt
srinivas@Srinivas:~/linux_task6$ mv important_backup.txt important_restored.txt
srinivas@Srinivas:~/linux_task6$ cat important_restored.txt
Line 1: Linux is powerful.
Line 2: File permissions are important.
Line 3: Backups prevent data loss.
Line 4: cp is used to copy files.
Line 5: mv is used to rename files.
Line 6: rm deletes files.
Line 7: Directories organize files.
Line 8: chmod changes permissions.
Line 9: Practice improves Linux skills.
Line 10: This is the last line.

srinivas@Srinivas:~/linux_task6$
```

Task9


```

srinivas@Srinivas:~$ mkdir linux_project
srinivas@Srinivas:~$ cd linux_project
srinivas@Srinivas:~/linux_project$ mkdir source backup docs
srinivas@Srinivas:~/linux_project$ ls
backup  docs  source
srinivas@Srinivas:~/linux_project$ touch source/main.c
srinivas@Srinivas:~/linux_project$ source/helper.c source/REANME.TXT
-bash: source/helper.c: No such file or directory
srinivas@Srinivas:~/linux_project$ touch source/main.c source/helper.c source/REANME.txt
srinivas@Srinivas:~/linux_project$ ls-R
ls-R: command not found
srinivas@Srinivas:~/linux_project$ ls -R
.:
backup  docs  source

./backup:

./docs:

./source:
REANME.txt  helper.c  main.c
srinivas@Srinivas:~/linux_project$ cat > source/README.txt
#!/bin/bash

# Define the output file name
OUTPUT_FILE="README.txt"

# Write the text lines directly to the file
cat << 'EOF' > "$OUTPUT_FILE"
Line 1: Linux Project
Line 2: Source files
Line 3: Main program
Line 4: Helper functions
Line 5: Documentation
Line 6: File management
Line 7: Backup process
Line 8: Copy command
Line 9: Move command
Line 10: Remove command
Line 11: Permissions
Line 12: Read access
Line 13: Write access
Line 14: Execute access
Line 15: Directories
Line 16: Practice commands
Line 17: Linux basics
Line 18: Terminal usage
Line 19: Project completed
Line 20: End of README

```

```

srinivas@Srinivas:~/linux_project$ head -5 source/README.txt
Line 1: Linux Project
Line 2: Source files
Line 3: Main program
Line 4: Helper functions
Line 5: Documentation
srinivas@Srinivas:~/linux_project$ tail -5/README.txt
tail: option used in invalid context -- 5
srinivas@Srinivas:~/linux_project$ tail -5 source/README.txt
Line 16: Practice commands
Line 17: Linux basics
Line 18: Terminal usage
Line 19: Project completed
Line 20: End of README
srinivas@Srinivas:~/linux_project$ cd source/main.c backup/
-bash: cd: too many arguments
srinivas@Srinivas:~/linux_project$ cp source/main.c backup/
srinivas@Srinivas:~/linux_project$ cp source/README.txt docs/
srinivas@Srinivas:~/linux_project$ mv source/helper.c source/functions.c
srinivas@Srinivas:~/linux_project$ chmod 644 source/README.txt
srinivas@Srinivas:~/linux_project$ rm backup/main.c
srinivas@Srinivas:~/linux_project$ rmdir backup
srinivas@Srinivas:~/linux_project$ tree
Command 'tree' not found, but can be installed with:
sudo snap install tree # version 2.1.3+pkg-5852, or
sudo apt install tree # version 2.3.1-1
See 'snap info tree' for additional versions.
srinivas@Srinivas:~/linux_project$ ls -R
.:
docs  source

./docs:
README.txt

./source:
README.txt  REANME.txt  functions.c  main.c
srinivas@Srinivas:~/linux_project$ |

```

Task10

```

srinivas@Srinivas:~$ mkdir student_portal
srinivas@Srinivas:~$ cd student_portal
srinivas@Srinivas:~/student_portal$ mkdir src docs backup longs config
srinivas@Srinivas:~/student_portal$ ls
backup  config  docs  longs  src
srinivas@Srinivas:~/student_portal$ touch src/main.c src/student.c docs/README.txt config/config.txt logs/system.log
touch: cannot touch 'logs/system.log': No such file or directory
srinivas@Srinivas:~/student_portal$ ls -R
.:
backup  config  docs  longs  src

./backup:

./config:
config.txt

./docs:
README.txt

./longs:

./src:
main.c  student.c
srinivas@Srinivas:~/student_portal$ cat > logs/system.log
-bash: logs/system.log: No such file or directory
srinivas@Srinivas:~/student_portal$ cat > longs/system.log\
>

Log Entry 1: System started.
Log Entry 2: User logged in.
Log Entry 3: Configuration loaded.
Log Entry 4: Database connected.
Log Entry 5: Student module initialized.
Log Entry 6: Login successful.
Log Entry 7: Data retrieved.
Log Entry 8: Backup completed.
Log Entry 9: User updated profile.
Log Entry 10: Session refreshed.
Log Entry 11: Security check passed.
Log Entry 12: New student added.
Log Entry 13: Log file updated.
Log Entry 14: System running normally.
Log Entry 15: End of log.
srinivas@Srinivas:~/student_portal$ head -5 longs/system.log

Log Entry 1: System started.
Log Entry 2: User logged in.
Log Entry 3: Configuration loaded.
Log Entry 4: Database connected.
srinivas@Srinivas:~/student_portal$ tail -5 longs/system.log
Log Entry 11: Security check passed.
Log Entry 12: New student added.
Log Entry 13: Log file updated.
Log Entry 14: System running normally.
Log Entry 15: End of log.
srinivas@Srinivas:~/student_portal$ cp docs/README.txt backup/
srinivas@Srinivas:~/student_portal$ mv backup/README.txt
mv: missing destination file operand after 'backup/README.txt'
Try 'mv --help' for more information.
srinivas@Srinivas:~/student_portal$ mv backup/README.txt backup/README_backup.txt

```

```
try mv --help for more information.
srinivas@Srinivas:~/student_portal$ mv backup/README.txt backup/README_backup.txt
srinivas@Srinivas:~/student_portal$ chmod 600 config/config.txt
srinivas@Srinivas:~/student_portal$ chmod 644 docs/README.txt
srinivas@Srinivas:~/student_portal$ rm backup/README_backup.txt
srinivas@Srinivas:~/student_portal$ ls -l
total 20
drwxr-xr-x 2 srinivas srinivas 4096 Aug  5 09:06 backup
drwxr-xr-x 2 srinivas srinivas 4096 Aug  5 08:59 config
drwxr-xr-x 2 srinivas srinivas 4096 Aug  5 08:59 docs
drwxr-xr-x 2 srinivas srinivas 4096 Aug  5 09:00 longs
drwxr-xr-x 2 srinivas srinivas 4096 Aug  5 08:59 src
srinivas@Srinivas:~/student_portal$ rmdir backup
srinivas@Srinivas:~/student_portal$ ls -R
.:
config docs longs src

./config:
config.txt

./docs:
README.txt

./longs:
system.long

./src:
main.c student.c
srinivas@Srinivas:~/student_portal$ |
```